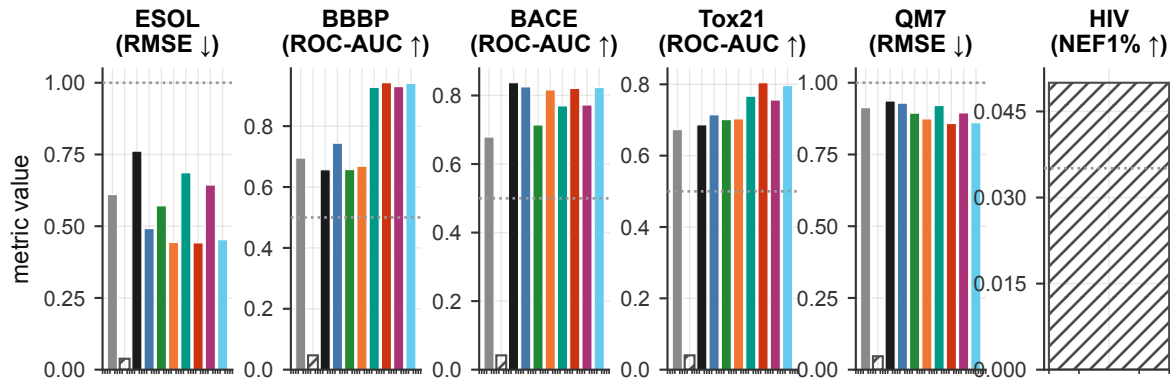


Fig A1 (H1) — headline comparison at 8M forward passes (single pretraining seed)

no_pretrain = random encoder, frozen, head on eval · no_pretrain_end_to_end (E1) = random encoder finetuned end-to-end · sup_only = random init → SFT (no MLM)



no_pretrain (frozen) unsup→sup (pending) sup_only: dense+sparse no_pretrain_end_to_end (E1, pending)
 Morgan+XGBoost (ECFP4) sup_only: dense (MTR) sup_only: minimol_full unsup→sup (not run)
 unsup_only (MLM) sup_only: sparse_all sup_only: mixed random (chance: 0.5 AUC / mean RMSE)